ALEXANDER SUBRAMANIAN

SOFTWARE ENGINEER

SKILLS

Languages: C#, C, C++, Python, Java, Bash, PowerShell

Libraries: .NET (WPF, Winforms, ASP.NET, OpenTk), Json.Net, Pandas, Numpy, Scikit-learn, Tensorflow, PyTorch, Matplotlib, Pandas`

Dev tools/tech: Visual Studio, VS Code, PyCharm, Anaconda, Docker, SOAP, REST, gRPC, NetBeans, Eclipse, Git, Mercurial, JIRA, Fisheye, ClearCase, ClearQuest, VAPS, DOORS, Fortify

Operating Systems: Windows (desktop and server), Linux (desktop, server, embedded distros), macOS, Android, iOS

EDUCATION

Arizona State University
Master of Computer Science
Expected; August 2021

University of Missouri
B.S. Electrical Engineering
Graduated: May 2012

PERSONAL PROFILE

Results-oriented Software Engineer proficient in all phases of the software development life cycle, including requirements definition, design, implementation, testing, and maintenance. Graduate school and hobbyist level experience with machine learning modeling using Python and C#. Experience working in classified environments and currently holding an active Secret DoD-issued security clearance.

WORK EXPERIENCE

SIERRA NEVADA CORPORATION – Denver, CO
Aerospace Software Engineer - Team Lead
JULY 2017 - PRESENT

- Software Team Lead of award-winning, classified training system used by multiple government agencies
- Integrated multiple new avionics, radios, and weapons systems onto the A-29 platform as part of the "Light Attack Experiment", ultimately winning a contract award
- Refactored legacy codebases and common code repositories to use updated libraries and improved architectural patterns
- Designed, developed, and demonstrated a prototype universal cockpit input device architecture
- Hire and manage a team of software engineers for a $24 million program and lead customer requirements meetings for development schedule planning
- Administrate JIRA projects, lead sprint planning, and oversee code reviews
- Develop and maintain system and network architecture

SECURITY CLEARANCE

DoD Top Secret SSBI
(Issued 1/11/2013)

CONTACT

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314-696-7216

THE BOEING COMPANY – St Louis, MO
F/A-18 Software Engineer
MAR 2015 - JUN 2017

- Designed and developed embedded software for the Advanced Mission Computers and front-end development for the cockpit displays of the F/A-18 and E/A-18G combat jets.
- Designed and implemented new cockpit display formats and features for the F/A-18 operation flight program to enhance flight crew situational awareness and mission capability
- Developed software interfacing multiple hardware components in a multi-layer software architecture requiring collaboration with multiple hardware vendors to implement unique transmission and messaging protocols
- Implemented scalable structural design patterns to provide reusable classes and software functionality for universal implementation on multiple hardware configurations

THE BOEING COMPANY – St Louis, MO
Phantom Works Electrophysics Engineer
JUN 2012 - MAR 2015

- Developed and maintained tools in C# and MATLAB that automate several data processing steps, saving engineering hours and increasing efficiency by 30%
- Performed data-reduction and modeling of complex data systems using C#, MATLAB, Excel, and other data analysis tools.
- Predicted electromagnetic properties of air vehicle components using method of moment and physical optics codes
- Presented and summarized radar cross-section and antenna gain analysis of surface features and antennas to an integrated product team in support of program level trade studies
- Collaborated with Computational Electromagnetics (CEM) code developers to implement and debug new features in the CEM tool suite